

YILI YANG, PhD

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Machine Learning for Earth Observation at Continental Scale

SKILLS

DOMAINS

Machine Learning & Deep Learning
Time-series modelling
Remote Sensing
Geospatial Information Science
Earth Foundation Models
Multimodal Learning
AI for Science
Scientific Data & Image Processing

TECH STACK

Python|PyTorch|Scikit-learn
Google Cloud Platform
Google Earth Engine

ENGINEERING

Docker|GitHub Actions
Agentic Coding
Prompt and Context Engineering
Git|GitHub

EDUCATION

UNIVERSITY OF EDINBURGH

PHD PETROPHYSICS, 2021
MSc GEOLOGY, 2016
BSc GEOSCIENCE, 2015

STATS

PUBLICATIONS

Journal Papers: 13
Conferences: 8
Data Sets: 1

ACADEMIC SERVICES

Invited Talks & Panels: 17
Peer-reviews: 8
Mentees: 9

LINKS

Personal Website
LinkedIn || ORCID
Google Scholar || GitHub

EXPERIENCE

DATA SCIENTIST | WOODWELL CLIMATE RESEARCH CENTER

Jan 2022 - Present | Falmouth, MA, USA

- **Project lead:** Pan-Arctic deep-learning programme detecting and quantifying permafrost thaw features from satellite imagery, feeding climate feedback and carbon-cycle research
- **Carbon:** Machine-learning imputation of carbon-flux time series; collaboration on permafrost thaw and carbon emission
- **Scale:** Optimised deep-learning inference over Arctic-scale geospatial data on Google Cloud Platform
- **Building for others:** Built a benchmark training dataset now used as a shared resource, and a zero-shot interactive segmentation workflow on the Segment Anything Model
- **Wildfire:** Deep-learning wildfire mapping from satellite imagery and machine-learning wildfire prediction from environmental variables
- **Working with others:** Ran concurrent modelling collaborations with three university research groups; lead the annual machine-learning workshop series and mentor interns

DATA SCIENCE FELLOW | FACULTY.AI

Sep 2021 - Dec 2021 | London, UK

- Commercial data-science fellowship in machine learning and AI, including an industry client placement

DOCTORAL RESEARCHER | ICCR

International Centre for Carbonate Reservoirs

Sep 2016 - Apr 2021 | Edinburgh, UK

- Industry-funded PhD developing computational methods for multiphase fluid flow in porous media
- Ran experiments at three national synchrotron facilities in the UK, US and Switzerland

SELECTED PUBLICATIONS

- Mapping retrogressive thaw slumps using deep neural networks — *Remote Sensing of Environment*, 2023.
- Data-driven modeling of carbon dioxide and methane fluxes across the Arctic-boreal region: recent achievements and future opportunities — *submitted to Global Change Biology*.
- A collaborative and scalable geospatial data set for arctic retrogressive thaw slumps with data standards — *Scientific Data*, 2025.

AWARDS & GRANTS

FUNDING FOR CLIMATE SOLUTIONS

Funding for Climate Solutions, Woodwell Climate Research Center
\$99,749 | 2025-2026 | A Generic Climate AI Framework for Multi-domain Time Series Prediction

MULTIDISCIPLINARY COLLABORATIONS

GOOGLE × WOODWELL

Understanding Permafrosts, AlphaEarth and Satellite Embedding

ESRI × WOODWELL

Understanding Permafrosts and Indigenous Communities